Notes

October 29, 2020

1. (a) *Proof.* Consider the coordinatization of $\mathbb{R}^2 - \{(0,0)\}$ using polar coordinates (r, θ), and recall that $x = r\cos(\theta)$ and $y = r\sin(\theta)$. Then, we obtain:

$$\begin{aligned} r^2 &= r^2(\cos^2(\theta) + \sin^2(\theta)) = x^2 + y^2 \\ \partial(r^2) &= 2r\partial(r) = 2x\partial(x) + 2y\partial(y) \\ \partial_r &= \frac{x\partial_x + y\partial_y}{x^2 + y^2}, \end{aligned}$$

and:

$$\begin{aligned} r\sin(\theta) + r\cos(\theta) &= y + x \\ \partial(\theta)(\cos(\theta) - \sin(\theta)) = \partial\left(\frac{y+x}{r}\right) &= \frac{r(\partial(y) + \partial(x)) - \partial(r)(y+x)}{r^2} = \frac{(x-y)(x\partial(y) - y\partial(x))}{(x-y)r^2} \\ \partial_\theta &= \frac{x\partial_y - y\partial_x}{x^2 + y^2} \end{aligned}$$

□

(b) *Proof.* Consider the coordinatization of $\mathbb{R}^3 - \{(0,0,0)\}$ using spherical coordinates $(\rho, \theta, \phi) \mapsto (x, y, z)$ where $x = \rho\sin\phi\cos\theta,\ y = \rho\sin\phi\sin\theta,\ z = \rho\cos\phi$.
Then, we obtain:

$$\begin{aligned} \partial(\rho^2) &= \partial(x^2 + y^2 + z^2) \\ \partial_\rho &= \frac{x\partial_x + y\partial_y + z\partial_z}{x^2 + y^2 + z^2}, \end{aligned}$$

and:

$$\begin{aligned} \arctan(\tan(\theta)) = \arctan\left(\frac{y}{x}\right) &= \arctan\left(\frac{\rho\sin\phi\cdot\sin\theta}{\rho\sin\phi\cdot\cos\theta}\right) \\ \partial_\theta &= \frac{1}{\left(\frac{y}{x}\right)^2 + 1} \cdot \frac{x\partial(y) - y\partial(x)}{x^2} = \frac{x\partial_y - y\partial_x}{x^2 + y^2}, \end{aligned}$$

and:

$$\begin{aligned} \partial\left(\arccos(\cos\phi)\right) &= \partial\left(\arccos\left(\frac{z}{\rho}\right)\right) \\ \partial_\phi &= \frac{-1}{\sqrt{1 - \left(\frac{z}{\rho}\right)^2}} \cdot \frac{\rho\partial(z) - z\partial(\rho)}{\rho^2} = \frac{xz\partial_x + yz\partial_y - (x^2+y^2)\partial_z}{(x^2+y^2+z^2)\sqrt{x^2+y^2}}. \end{aligned}$$

□

2. *Proof.* Applying Exercise 10(a) yields:

$$\begin{aligned} ds^2 = (dx)^2 + (dy)^2 &= \left(\frac{\partial x}{\partial r}dr + \frac{\partial x}{\partial\theta}d\theta\right)^2 + \left(\frac{\partial y}{\partial r}dr + \frac{\partial y}{\partial\theta}d\theta\right)^2 \\ &= (\cos\theta dr - r\sin\theta d\theta)^2 + (\sin\theta dr + r\cos\theta d\theta)^2 = (\cos^2\theta + \sin^\theta)dr^2 + 0 + (r^2\sin^2\theta + r^2\cos^2\theta)d\theta^2 \\ &= dr^2 + r^2 d\theta^2. \end{aligned}$$

□

3. Consider the polar coordinates (r, θ) in the Euclidean place, consider the line element $ds^2 = dr^2 + f(r)^2 d\theta^2$ for some positive function $f(r)$, and consider some curve $\gamma(t) = (r(t), \theta(t))$.

   (a) For a fixed value of $\theta_\star$, we compute the length of $\gamma(t) = (t, \theta_\star),\ 0 < t < \rho$ below:

   $$ds^2 = (r(\rho) - r(0))^2 + f(r)^2 \cdot 0^2 = \rho^2.$$

   (b) For a fixed value of ρ, we compute the length of $\gamma(t) = (\rho, t),\ -\pi < t < \pi$ in terms of f below:

   $$ds^2 = 0^2 + f(\rho)^2 \left(\pi - (-\pi)\right)^2 = 4f(\rho)^2\pi^2.$$

   (c) Consider the following choices of f.

      i. $f(r) = r$. This is the usual Euclidean geometry viewed in polar coordinates $(r, \theta) \mapsto (r\cos\theta, r\sin\theta)$.

      ii. $f(r) = \sin(r)$ under $0 < r < \pi$. This is also the usual Euclidean geometry viewed in spherical coordinates $x = \sin(r)\cos(\theta),\ y = \sin(r)\sin(\theta)$ where r plays the usual role of ϕ, and where we ignore the z-coordinate.

      iii. $f(r) = 1$. This corresponds to the unit circle.

   (d) i. $f(r) = \frac{1}{2}r$. This looks like the Euclidean plane viewed with the usual polar coordinates, but with distances distorted half as much.

      ii. $f(r) = 2r$. As above but distances are distorted twice as much.

      iii. $f(r) = 1 + 2r$.

      iv. $f(r) = 2 + \sin(r)$.

      v. $f(r) = e^r$. This looks like the usual Euclidean plane with the usual polar coordinates, but with distances distorting exponentially as we venture out.

      vi. $f(r) = \sinh(r) = \frac{1}{2}(e^r - e^{-r})$. Since we only consider positive r, this looks very much like the geometry described above for large r.

0.1 Agenda

Geometry, Courses, Teaching, Good of the Order